

Team Name : **JugadX**

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Problem Statement : **PS-06 - AI-based Automated Crop Type & Moisture Stress Detection using Optical & Microwave Satellite Data**

Link to GitHub Repository: https://github.com/manasmmm/PAM-ROFI

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How is it different?

Multi-Sensor Fusion

Most crop-mapping tools use only a single sensor and stop at a static classification map.

We fuse **Sentinel-2 optical + Sentinel-1 SAR** and a phenology-aware time token into two purpose-built frameworks (PAM and Tabular PAM).

How it Solves it

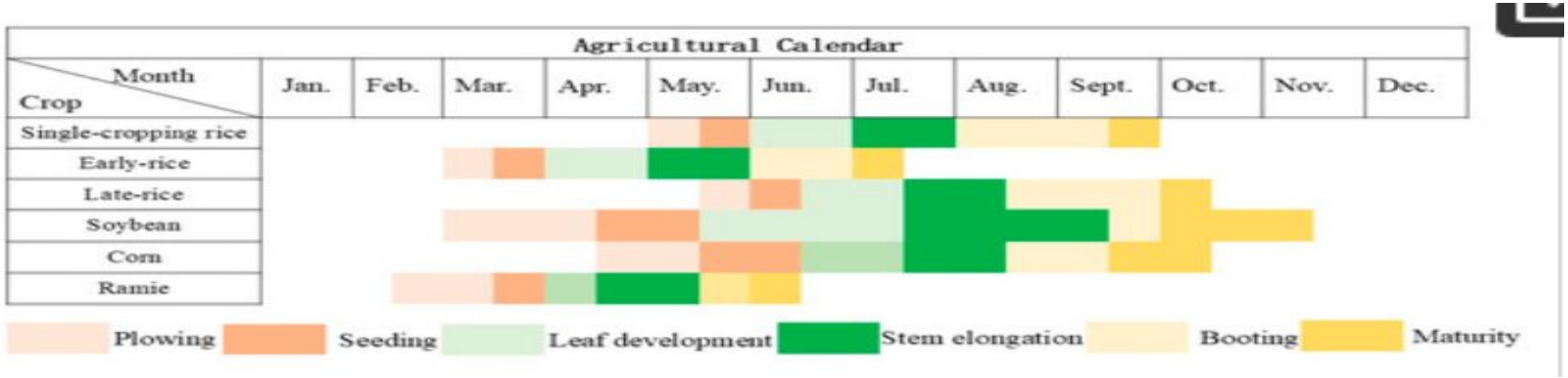
Automated Pipeline

Runs entirely on free satellite data via GEE. It classifies **4 crop/land classes**, flags moisture stress at **8-day resolution**, and issues a 3-tier irrigation advisory with cloud-resilient fallback.

Our USP

First-Class Phenology

Treating phenology as a tabular feature. Crop-calendar context (PAM tokens) is injected into a lightweight Random Forest, giving **deep-learning-grade awareness** without GPU clusters.



Key Features of the Solution

01 Multi-Sensor Fusion

Sentinel-1 SAR + Sentinel-2 optical, cloud-masked and enriched with NDVI, EVI, NDWI and SAR ratio.

02 Phenology-Aware Compositing

10-day composites tagged with a crop growth-stage (PAM) token across the Kharif season.

03 4-Class Crop Classifier

Random Forest model reaching 94.0% accuracy and a kappa of 0.85 on held-out validation.

04 Moisture Stress Detection

VCI + SMI phase-aware decision matrix flags stress at 8-day resolution.

05 Irrigation Advisory Engine

FAO-56 Kc curves drive ETc, water deficit, and a 3-tier irrigation advisory.

06 Time-Series Inspector

Click any point on the map to generate a live moisture-stress trajectory chart.

07 Cloud-Resilient Fallback

Auto-switches to a 30-day window whenever the 8-day window has no clear imagery.

```
=== GENERATING PHENOLOGY-BASED CROP TARGET GROUND TRUTH ===

--- MEERUT LOCAL 70/30 VALIDATION RESULTS ---
Confusion Matrix Array:
 [[0, 0, 0, 0], [0, 26, 10, 0], [0, 0, 1065, 15], [0, 0, 42, 331]]
Overall Validation Accuracy: 95.50%
Kappa Coefficient Check: 0.8867
```

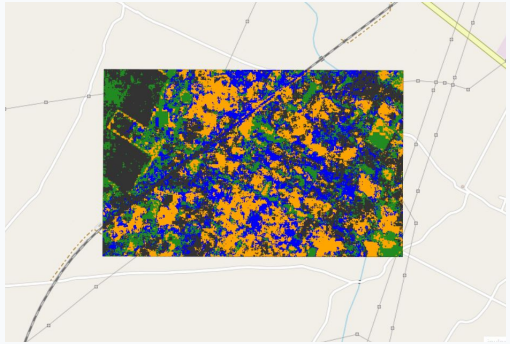
End-to-End Process Flow

1. Inputs	2. Preprocessing	3. Processing	4. Output Layer
Sentinel-2 Optical · 10m	**Cloud Masking** QA60 + SCL band	**SAR Fused Stack** S2 indices + S1 VV/VH + PAM	**Crop Type Map** 4 classes · OA target >85% + Kappa
Sentinel-1 SAR · C-band	**Index Computation** NDVI · EVI · NDWI	**Random Forest** 100 trees · 70/30 split	
Landsat 8/9 Thermal band	***PAM Token** Stage 0-3 per step		
MOD16A2 PET · 8-day	**SAR Speckle Filter** VV · VH · ratio	**VCI + SMI Baselines** Min/max envelopes	**Stress Layer** 8-day stage-aware · no/mild/severe
CHIRPS + GFS Rainfall + forecast	**10-day Composites** Regularised S1 + S2	**Stage-Aware Engine** PAM × VCI × SMI	
SMAP Soil moisture ref.		**PM-ETc Calculation** ET0 × Kc (FAO-56)	
FAO-56 Kc Calendar Stage + sowing date	**SEBAL ET Extraction** Thermal → actual ET	***NWR Water Balance** PM-ETc - SEBAL - SM	**Advisory Map** Field + canal dispatch · 8-day

OUR INNOVATION-PAM
***Phenology Alignment Method**

Our model replaces complex **spatial-temporal Transformers** with a lean, physics-grounded alternative. Instead of using heavy neural networks to blindly calculate attention weights, our engine injects metadata-driven Tabular PAM tokens directly into a fused Sentinel optical-radar data stack. This hands explicit lifecycle context to a ***Random Forest*** model, matching ***deep-learning precision*** with instant cloud-server efficiency

Live Outputs from the Working Pipeline (Meerut Pilot)



4-Class Crop Map

Daurala Pilot Plot Classification:

- **Class 1:** Non-agri (grey)
- **Class 2:** Sugarcane (green)
- **Class 3:** Paddy rice (blue)
- **Class 4:** Wheat/Fallow (yellow)



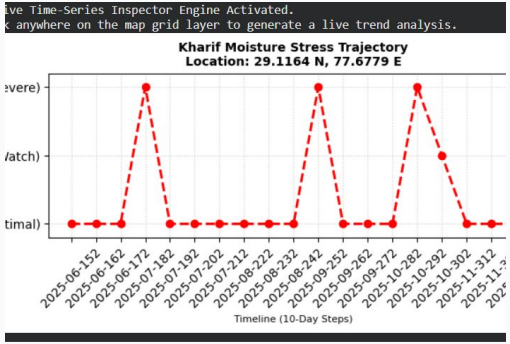
Moisture Stress

Real-time stress levels:

Level 1: No stress (green)

Level 2: Mild depletion (orange)

Level 3: Severe deficit (red)



Interactive Inspector

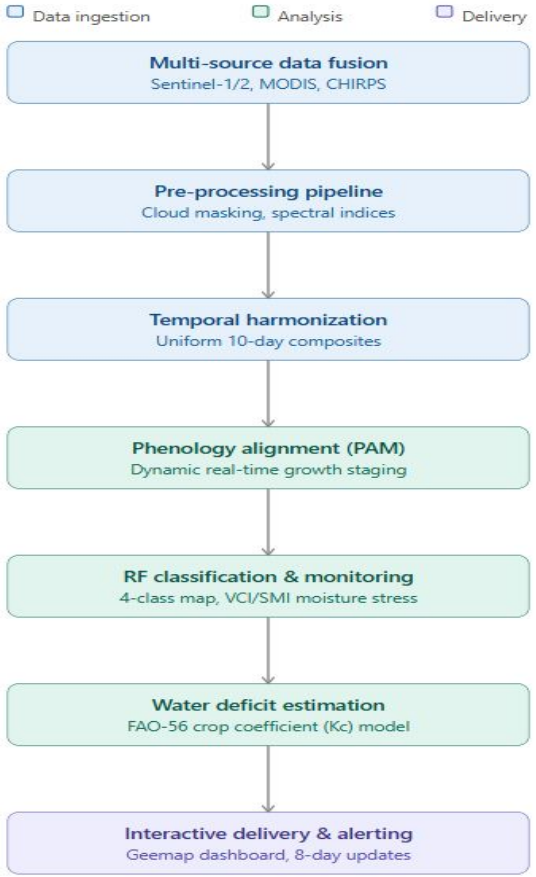
Time Series Engine:

Click anywhere on the map grid layer to generate a live trend analysis.

- **Instant plotting of data**
- **Multi-temporal tracking**

System Architecture

- **Multi-Source Data Fusion:** Integrates four satellite streams over the **Meerut** pilot region via **Google Earth Engine**: Sentinel-1 (SAR), Sentinel-2 (Optical), MODIS (Evapotranspiration), and CHIRPS (Rainfall).
- **Pre-Processing Pipeline:** Executes cloud masking using the **QA60 band**, followed by computation of spectral indices (**NDVI, EVI, NDWI**) and the **SAR VV/VH** ratio.
- **Temporal Harmonization:** Resamples and processes all datasets into uniform, regular **10-day composites** to ensure consistency.
- **Phenology Alignment (PAM):** The **Phenology Alignment Method** module analyzes integrated temporal signatures to assign real-time growth stages dynamically, bypassing rigid calendar dates.
- **Random Forest Classification:** Fuses spatial features to map four distinct classes: **Non-Agricultural land, Sugarcane, Paddy, and Wheat/Fallow** using adaptive local thresholds.
- **Moisture Stress Monitoring:** Leverages a decision matrix based on the **Vegetation Condition Index (VCI)** and **Soil Moisture Index (SMI)** from historical records.
- **Water Deficit Estimation:** Applies the standard **FAO-56 crop coefficient (Kc)** model, interpolating curves by crop type and days after sowing to evaluate real deficits against rainfall.
- **Interactive Delivery & Alerting:** Visualizes three advisory levels on a split-panel **geemap**, offering click-based time-series data and cloud-resilient **8-day updates** for field officers.



Technology Stack

Data & Sensors

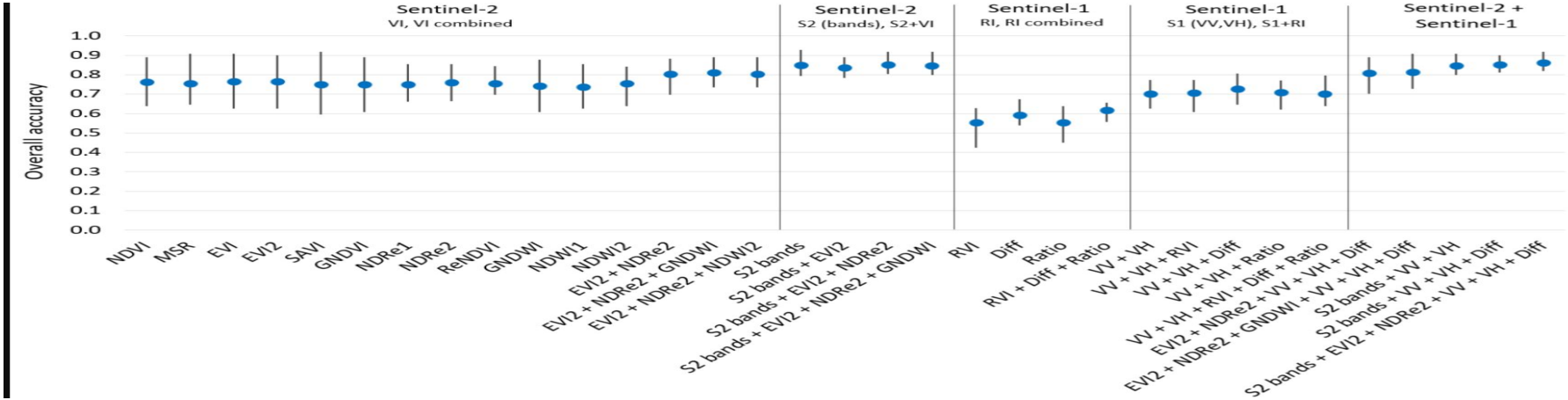
Sentinel-1 GRD (SAR)
Sentinel-2 SR (Optical)
MODIS MOD16A2GF (ET)
CHIRPS & GFS Forecasts

Processing & Modeling

GEE & geemap
Python
Random Forest Classifier
FAO-56 Kc Interpolation

Visualization & Delivery

Matplotlib + ipywidgets
Split-panel Map Viewer
Click-to-Chart UI
Dashboard System



Estimated Implementation Cost

The proposed system is designed to be cost-effective by leveraging freely available satellite datasets and cloud-native processing platforms **(Google Earth Engine)**. Its modular architecture minimizes development and maintenance overhead while enabling seamless scalability from pilot deployments to large-scale implementation, making it well suited for both research applications and real-world agricultural decision-support systems.

~ All the additional analysis and implementation has been documented and compiled in a supporting document and a python notebook, all of which has been added to a GitHub repo, the link for which is given in the first slide.


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THANK YOU